



**Generation Interconnection
Impact Study Report
for
Queue Project AE2-040
SAPONY 34.5 KV
15.7 MW Capacity / 15.7 MW Energy**

Revision 2: December 2021

Revision 1: December 2021

Revision 0: February 2020

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1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Feasibility Study Agreement between Shands Energy Center, LLC, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

3 Revision Summary

3.1 Revision 1 Summary – December 2021

This revision is being issued to incorporate the results of a re-tool.

3.2 Revision 2 summary – December 2021

This revision is being issued to include a supplemental upgrade instead of the previously identified network upgrade. The scope of the network upgrade previously identified is covered under s2610. Therefore, this project does not require a separate network upgrade and can take credit for the work being performed under s2610 to resolve the identified overload.

4 General

The Interconnection Customer (IC), has proposed a Storage generating facility located in Sussex County, Virginia. The installed facilities will have a total capability of 15.7 MW with 15.7 MW of this output being

recognized by PJM as Capacity. The proposed in-service date for this project is 06-01-2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-040
Project Name	SAPONY 34.5 KV
Interconnection Customer	Shands Energy Center, LLC
State	VA
County	Sussex
Transmission Owner	Dominion
MFO	15.7
MWE	15.7
MWC	15.7
Fuel	Storage
Basecase Study Year	2022

4.1 Point of Interconnection

AE2-040 will interconnect with the Dominion distribution system at Sapony 34.5 kV bus fed by Dominion's 230kV system. Attachment 1 shows a one-line diagram of the proposed interconnection facilities. The IC may not install any facilities on Dominion's right-of-way without first obtaining the necessary approval from Dominion Energy.

4.2 Cost Summary

The AE2-040 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ N/A
Direct Connection Network Upgrade	\$ N/A
Non Direct Connection Network Upgrades	\$ N/A
Total Costs	\$ N/A

Attachment facilities and local upgrades (if required) along with terms and conditions to interconnect AE2-040 will be specified in a separate two party Interconnection Agreement (IA) between ITO and the IC as this project is considered FERC non-jurisdictional per the PJM Open Access Transmission Tariff (OATT). The single line is shown below in Attachment 1.

In addition, the AE2-040 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$0

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not

closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade

5 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AE2-040 was evaluated as a 15.7 MW Capacity (15.7 MW energy) load reducer at the 34.5 kV Sapony substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2022 AE2 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically in Planning Studies NERC Category C Contingency Conditions (Bus Fault, Tower Line, N-1-1, and Stuck Breaker scenarios) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

Note that the ITO findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in a future study phases. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

6 Non-Direct Connection Cost Estimate

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

7 Schedule

The schedule of work by Dominion to interconnect the IC will be specified in a separate two party Interconnection Agreement (IA) between ITO and the IC as this project is considered FERC non-jurisdictional per the PJM Open Access Transmission Tariff (OATT).

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2022 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system. The AE2-040 project contributes to overloads on the Dominion transmission system as shown in the “Network Impact” section of the report. The estimated cost of system reinforcements necessary to mitigate these overloads is also provided.

8.2 Short Circuit Analysis

PJM performed a short circuit analysis and the results were verified by Dominion. The connection of AE2-040 project to the system does not result in any newly overdutied circuit breakers on the Dominion transmission system and does not have a significant fault current contribution to existing overdutied circuit breakers

8.3 Stability Analysis

Not Required.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion's "Dominion Energy Electric Transmission Generator Interconnection Requirements" documented in Dominion's Facility Interconnection Requirements "Exhibit C" located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC's GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Dominion Requirements

See Section 3.4.6 “Metering and Telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

11 Network Impacts

The Queue Project AE2-040 was evaluated as a 15.7 MW (Capacity 15.7 MW) injection at the 115 kV bus 3 69T148 in the Dominion area. Project AE2-040 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-040 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

12 Summer Peak Load Flow Analysis

12.1 Generation Deliverability – Summer Peak Load

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

Contingency			Affected Area	Facility Description	Bus			Power Flow	Loading %		Rating Type	MVA	MW Contribution	Flowgate Appendix
#	Type	Name			From	To	Ckt		Initial	Final				
1	N-1	DVP_P1-2: LN 100_AE2-027A	DVP - DVP	6CHESTFB-6BASIN 230 kV line	314287	314276	1	AC	99.94	100.53	ER	449	2.65	1
2	N-1	DVP_P1-3: 6CHESTFA-TX#9	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	99.3	102.14	ER	138	3.99	

12.2 Multiple Facility Contingency – Summer Peak Load

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

12.3 Contribution to Previously Identified Overloads – Summer Peak Load

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Contingency			Affected Area	Facility Description	Bus			Power Flow	Loading %		Rating Type	MVA	MW Contribution	Flowgate Appendix
#	Type	Name			From	To	ckt		Initial	Final				
3	LFFB	DVP_P4-2: 562T563	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	115.45	117.95	ER	169	4.32	2
4	LFFB	DVP_P4-6: T672	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	110.22	112.71	ER	169	4.25	
5	LFFB	DVP_P4-5: L642	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	105.29	107.81	ER	169	4.33	
6	BUS	DVP_P2-2: BASIN B7	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	105.25	107.75	ER	169	4.33	
7	LFFB	DVP_P4-2: 206542	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	105.25	107.75	ER	169	4.33	
8	LFFB	DVP_P4-2: 25942	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	105.25	107.75	ER	169	4.33	
9	DCTL	DVP_P7-1: LN 259-2065	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	103.79	106.31	ER	169	4.36	
10	DCTL	DVP_P7-1: LN 259-282	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	103.36	105.88	ER	169	4.36	

12.4 Potential Congestion due to Local Energy Deliverability – Summer Peak Load

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection

Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

Overload		Contingency			Bus				Loading %		Rating		MW	Flowgate
Number	Type	Name	Affected Area	Facility Description	From	To	Circuit	Power Flow	Initial	Final	Type	MVA	Contribution	Appendix
11	N-1	DVP P1-2: LN 259	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	124.8	127.9	ER	138	4.39	
12	N-0	Basecase	DVP - DVP	AE2-027 TAP-3HARROWG 115 kV line	940430	314298	1	AC	109.09	112.1	NR	138	4.27	

12.5 System Reinforcements – Summer Peak Load

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE2-040 Allocation
#1	6CHESTF B-6BASIN 230 kV line	Chesterfield to Basin 230 kV line - Replace 0.14 miles of 1109 ACAR with a conductor which will increase the line rating to approximately 706 MVA. Note: Project is in-service as of 4/27/2018 and the rating is sufficient for this project.	b2990	\$350,000	\$0
#2	AE2-027 TAP-3HARROWG 115 kV line	AE2-040 is the driver for the overload of this flowgate, and also a contributor to a previously identified overload for the same monitored facility. See the “Contribution to Previously Identified System Reinforcements” section below for the reinforcement and cost allocation, if any.	-	-	-
Total Cost(\$)				\$350,000	\$0

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE2-040 Allocation
#3, 4, 5, 6, 7, 8, 9, 10	AE2-027 TAP-3HARROWG 115 kV line	<u>DVP Supplemental Project:</u> Project Id: s2610 Project Description: Rebuild all wood H frame structures on 115kV Line Locks-Chesterfield from Locks to Harrowgate and reconductor the 5.4 miles with current 115 kV standards construction practices. Upgrade terminal equipment as needed. The normal summer rating of the line will be 393 MVA. Project Type: FAC Cost: \$6,900,000 Projected ISD: 12/31/2022	s2610	\$6,900,000	\$0
Total Cost(\$)				\$6,900,000	\$0

13 Flow Gate Details – Summer Peak Load

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

13.1 Index 1

(DVP - DVP) The 6CHESTF B-6BASIN 230 kV line (from bus 314287 to bus 314276 ckt 1) loads from 99.94% to 100.53% (AC power flow) of its emergency rating (449 MVA) for the single line contingency outage of 'DVP_P1-2: LN 100_AE2-027A'. This project contributes approximately 2.65 MW to the thermal violation.

CONTINGENCY 'DVP_P1-2: LN 100_AE2-027A'

OPEN BRANCH FROM BUS 314284 TO BUS 314344 CKT 1 /* 3CHESTFLD 115.00 - 3TYLER 115.00
 OPEN BRANCH FROM BUS 314298 TO BUS 940430 CKT 1 /* 3HARROWG 115.00 - 3LOCKS 115.00
 OPEN BRANCH FROM BUS 314298 TO BUS 314349 CKT 1 /* 3HARROWG 115.00 - 3WALTHAL 115.00
 OPEN BRANCH FROM BUS 314344 TO BUS 314349 CKT 1 /* 3TYLER 115.00 - 3WALTHAL 115.00
 OPEN BUS 314298 /* ISLAND: 3HARROWG 115.00
 OPEN BUS 314344 /* ISLAND: 3TYLER 115.00
 OPEN BUS 314349 /* ISLAND: 3WALTHAL 115.00
 END

Bus Number	Bus Name	Full Contribution
315065	1CHESTF6	26.16
315139	1GASTONA	1.02
315141	1GASTONB	1.02
315119	1GRAVEL3	0.83
315120	1GRAVEL4	0.84
315121	1GRAVEL5	0.83
315122	1GRAVEL6	0.84
315117	1GRAVELC	0.29
315074	1HOPCGN1	5.63
315075	1HOPCGN2	5.56
315077	1HOPHCF1	1.56
315078	1HOPHCF2	1.56
315079	1HOPHCF3	1.56
315080	1HOPHCF4	2.37
315076	1HOPPOLC	1.
315136	1ROSEMG1	0.7
315138	1ROSEMG2	0.33
315137	1ROSEMS1	0.44
315116	1SURRY1	8.14
314704	3LAWRENC	0.11
314314	3LOCKS	0.26
932591	AC2-079 C O1	2.29
933261	AC2-137 C	-0.64
934011	AD1-025 C	8.24
934571	AD1-082 C	4.

935161	AD1-151 C O1	8.28
936041	AD2-007	0.87
936051	AD2-008 C	1.43
936661	AD2-085 C	1.4
938631	AE1-085 C O1	6.14
939191	AE1-149 C O1	8.05
939311	AE1-162 C	0.63
940431	AE2-027 C O1	12.16
940481	AE2-033 C	9.11
940541	AE2-040 O1	2.65
940651	AE2-052	2.68
940891	AE2-078 C	0.75
940901	AE2-079 C	0.75
941531	AE2-150	0.12
942001	AE2-212 C	2.53
942161	AE2-228 C	2.87
942341	AE2-247 C	0.61
942371	AE2-250 C O1	9.12
942471	AE2-260 C O1	6.93
942552	AE2-270 B C	5.32
942551	AE2-270 S C	7.98
LTF	CARR	0.19
LTF	CBM-S1	1.81
LTF	CBM-S2	2.6
LTF	CBM-W1	1.2
LTF	CBM-W2	11.2
LTF	CIN	0.58
LTF	CPL	1.41
LTF	IPL	0.35
LTF	LGEE	0.17
LTF	MEC	1.5
LTF	MECS	0.26
LTF	RENSSELAER	0.15
LTF	WEC	0.15
930861	AB1-132 C O1	5.03
931231	AB1-173 C	0.83
931241	AB1-173AC	0.83
923851	AB2-025 C	0.23
923911	AB2-031 C O1	0.82
923991	AB2-040 C O1	2.71
924511	AB2-100 C	5.43
924811	AB2-134 C O1	0.97
925051	AB2-160 C O1	5.13
925061	AB2-161 C O1	1.76
925171	AB2-174 C O1	2.67

925331	<i>AB2-190 C</i>	<i>10.3</i>
925821	<i>AC1-061</i>	<i>< 0.01</i>
926071	<i>AC1-086 C</i>	<i>7.41</i>
927221	<i>AC1-216 C O1</i>	<i>4.79</i>

13.2 Index 2

(DVP - DVP) The AE2-027 TAP-3HARROWG 115 kV line (from bus 940430 to bus 314298 ckt 1) loads from 115.45% to 117.95% (AC power flow) of its load dump rating (169 MVA) for the line fault with failed breaker contingency outage of 'DVP_P4-2: 562T563'. This project contributes approximately 4.32 MW to the thermal violation.

CONTINGENCY 'DVP_P4-2: 562T563' /* CARSON 500 KV
 OPEN BRANCH FROM BUS 314902 TO BUS 940640 CKT 1 /* 8CARSON 500.00 - 8SEPTA
 500.00
 OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 -
 8MDLTHAN 500.00
 END

Bus Number	Bus Name	Full Contribution
314314	3LOCKS	0.42
938631	AE1-085 C O1	2.85
938632	AE1-085 E O1	1.42
939191	AE1-149 C O1	3.78
939192	AE1-149 E O1	2.52
940431	AE2-027 C O1	35.98
940432	AE2-027 E O1	23.99
940541	AE2-040 O1	4.32
940651	AE2-052	1.26
941531	AE2-150	0.06
942001	AE2-212 C	0.71
942002	AE2-212 E	0.47
942371	AE2-250 C O1	14.87
942372	AE2-250 E O1	7.85
LTF	CARR	0.09
LTF	CBM-S1	1.47
LTF	CBM-S2	1.74
LTF	CBM-W1	1.39
LTF	CBM-W2	9.53
LTF	CIN	0.64
LTF	CPL	0.9
LTF	G-007	0.29
LTF	IPL	0.4
LTF	LGEE	0.18
LTF	MEC	1.44
LTF	MECS	0.55
LTF	O-066	1.83
LTF	RENSSELAER	0.07
LTF	WEC	0.17
925051	AB2-160 C O1	8.37
925052	AB2-160 E O1	13.66
925821	AC1-061	< 0.01

14 Contingency Details – Summer Peak Load Flow

Contingency Name	Description
DVP_P1-2: LN 100_AE2-027A	CONTINGENCY 'DVP_P1-2: LN 100_AE2-027A' OPEN BRANCH FROM BUS 314284 TO BUS 314344 CKT 1 /* 3CHESTFLD 115.00 - 3TYLER 115.00 OPEN BRANCH FROM BUS 314298 TO BUS 940430 CKT 1 /* 3HARROWG 115.00 - 3LOCKS 115.00 OPEN BRANCH FROM BUS 314298 TO BUS 314349 CKT 1 /* 3HARROWG 115.00 - 3WALTHAL 115.00 OPEN BRANCH FROM BUS 314344 TO BUS 314349 CKT 1 /* 3TYLER 115.00 - 3WALTHAL 115.00 OPEN BUS 314298 /* ISLAND: 3HARROWG 115.00 OPEN BUS 314344 /* ISLAND: 3TYLER 115.00 OPEN BUS 314349 /* ISLAND: 3WALTHAL 115.00 END
DVP_P1-2: LN 259	CONTINGENCY 'DVP_P1-2: LN 259' OPEN BRANCH FROM BUS 314276 TO BUS 314287 CKT 1 /* 6BASIN 230.00 - 6CHESTF B 230.00 END
DVP_P1-3: 6CHESTF A-TX#9	CONTINGENCY 'DVP_P1-3: 6CHESTF A-TX#9' OPEN BRANCH FROM BUS 314284 TO BUS 314286 CKT 1 /* 3CHESTFLD 115.00 - 6CHESTF A 230.00 END
DVP_P2-2: BASIN B7	CONTINGENCY 'DVP_P2-2: BASIN B7' /* BASIN 230 KV OPEN BRANCH FROM BUS 314276 TO BUS 314287 CKT 1 /* 6BASIN 230.00 - 6CHESTF B 230.00 OPEN BRANCH FROM BUS 314276 TO BUS 314339 CKT 1 /* 6BASIN 230.00 - 6SPRUNCE 230.00 OPEN BRANCH FROM BUS 314274 TO BUS 314276 CKT 2 /* 3BASIN 115.00 - 6BASIN 230.00 END

DVP_P4-2: 206542	CONTINGENCY 'DVP_P4-2: 206542' /* BASIN 230 KV	
	OPEN BRANCH FROM BUS 314276 TO BUS 314339 CKT 1	/* 6BASIN 230.00 - 6SPRUNCE 230.00
	OPEN BRANCH FROM BUS 314339 TO BUS 315085 CKT 1	/* 6SPRUNCE 230.00 - 1SPRUNCC 13.800
	OPEN BRANCH FROM BUS 314339 TO BUS 315086 CKT 1	/* 6SPRUNCE 230.00 - 1SPRUNCD 13.800
	OPEN BUS 315085	/* ISLAND: 1SPRUNCC 13.800
	OPEN BUS 315086	/* ISLAND: 1SPRUNCD 13.800
	OPEN BRANCH FROM BUS 314276 TO BUS 314287 CKT 1	/* 6BASIN 230.00 - 6CHESTF B 230.00
	OPEN BRANCH FROM BUS 314274 TO BUS 314276 CKT 2	/* 3BASIN 115.00 - 6BASIN 230.00
	END	
DVP_P4-2: 25942	CONTINGENCY 'DVP_P4-2: 25942' /* BASIN 230 KV	
	OPEN BRANCH FROM BUS 314276 TO BUS 314287 CKT 1	/* 6BASIN 230.00 - 6CHESTF B 230.00
	OPEN BRANCH FROM BUS 314276 TO BUS 314339 CKT 1	/* 6BASIN 230.00 - 6SPRUNCE 230.00
	OPEN BRANCH FROM BUS 314274 TO BUS 314276 CKT 2	/* 3BASIN 115.00 - 6BASIN 230.00
	END	
DVP_P4-2: 562T563	CONTINGENCY 'DVP_P4-2: 562T563' /* CARSON 500 KV	
	OPEN BRANCH FROM BUS 314902 TO BUS 940640 CKT 1	/* 8CARSON 500.00 - 8SEPTA 500.00
	OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1	/* 8CARSON 500.00 - 8MDLTHAN 500.00
	END	
DVP_P4-5: L642	CONTINGENCY 'DVP_P4-5: L642' /* BASIN 115 KV	
	OPEN BRANCH FROM BUS 314274 TO BUS 314319 CKT 1	/* 3BASIN 115.00 - 3MAURY24 115.00
	OPEN BRANCH FROM BUS 314274 TO BUS 314324 CKT 1	/* 3BASIN 115.00 - 3PIPELIN 115.00
	OPEN BRANCH FROM BUS 314276 TO BUS 314287 CKT 1	/* 6BASIN 230.00 - 6CHESTF B 230.00
	OPEN BRANCH FROM BUS 314276 TO BUS 314339 CKT 1	/* 6BASIN 230.00 - 6SPRUNCE 230.00

OPEN BUS 314311

/* ISLAND: 6IRON282 230.00

OPEN BUS 315083

/* ISLAND: 1SPRUNCA 13.800

OPEN BUS 315084

/* ISLAND: 1SPRUNCB 13.800

END

Light Load Analysis

15 Light Load Load Flow Analysis

15.1 Generation Deliverability – Light load

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

15.2 Multiple Facility Contingency – Light load

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

15.3 Contribution to Previously Identified Overloads – Light load

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

15.4 Potential Congestion due to Local Energy Deliverability – Light load

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

15.5 System Reinforcements – Light load

None

16 Flow Gate Details – Light Load

None

17 Contingency Details – Light Load

None

Affected Systems

18 Affected Systems

None.

Short Circuit

19 Short Circuit

The following Breakers are overdutied:

None.

Stability

20 Stability

Not Required.

Attachment 1 – Single Line Diagram

